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Exercise 1 — Student Portal Navigation System

Topic: React Router – Simple Routing

Scenario

A university wants to create a Student Portal with:

- Home Page
- Student List Page
- Reports Page

The navigation should happen without page refresh.

Task

1. Create routing and redirect to students
2. Store the username
3. Display current route inside an `<Alert>`
4. Create responsive navigation using `<Navbar>`

State Management Requirement - Maintain:

- Logged-in username
- Current selected page
- Navigation history count

React Bootstrap Components need to be used

- `<Navbar>`, `<Nav>`, `<Container>`, `<Button>`, `<Alert>`

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Exercise 2 — Dynamic Student Profile Viewer

Topic: Route Parameters

Scenario

A college management system should display student details dynamically using Student ID.

Example:

/student/101

/student/102

Task

1. Create student records array
2. Navigate using links
3. Fetch student data
4. Display details inside **<Card>**
5. Add “Next Student” button

Task

1. Create student records array
2. Navigate using links
3. Fetch student data
4. Display details inside **<Card>**
5. Add “Next Student” button

State Management Requirement- Maintain:

- Selected student details
- Student marks
- Current student ID

React Bootstrap Components need to be used

- **<Card>** , **<Card.Body>** , **<Card.Title>**, **<Card.Text>** , **<Button>**

Sample URL

<http://localhost:3000/student/101>

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Exercise 3 — Product Search and Filter Dashboard

Topic: Query String Routing

Scenario

- An e-commerce application allows users to search via Query Strings:
 - Search products
 - Filter by category
 - Sort products

Example:

/products?category=laptop&sort=price

Task

1. Create product search form
2. Update URL using query strings
3. Read query values
4. Display current filters inside **<Alert>**
5. Preserve filters after browser refresh

State Management Requirement-Maintain:

- Search text
- Selected category
- Sorting option

React Bootstrap Components need to be used

- **<Form>** , **<Form.Group>** , **<Form.Label>** , **<Form.Control>** , **<Button>**,
<Alert>

Task

6. Create product search form
7. Update URL using query strings
8. Read query values using `useSearchParams()`
9. Display current filters inside **<Alert>**
10. Preserve filters after browser refresh

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Exercise 4 — Employee Status Filter System

Topic: Programmatic Navigation

Scenario

A company HR dashboard allows filtering employees by:

- Active
- Inactive
- On Leave

The filter should automatically change the URL dynamically.

Task

1. Create dropdown filter
2. Use `navigate()` on selection change
3. Update: `/employees?status=active`
4. Display filtered employees in Bootstrap table
5. Show success message using `<Alert variant="success">`

State Management Requirement

Maintain:

- Employee status
- Filtered employee list
- Current query string

React Bootstrap Components need to be used

- `<Form.Select>`, `<Table>`, `<Alert>` , `<Button>`

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Exercise 5 — Online Learning Dashboard

Topic: Nested Routes

Scenario

An online learning platform has:

- Courses Page
- Search Section
- Course List Section

The left navigation panel must remain visible while content changes dynamically.

1. Create parent route `/courses`
2. Create child routes:
 - `/courses/search`
 - `/courses/list`
3. Use `<Outlet />`
4. Build split-screen dashboard layout
5. Show selected course inside `<Card>`

State Management Requirement

Maintain:

- Selected course
- Current child route
- Sidebar collapse state

React Bootstrap Components need to be used

- `<Navbar>`, `<Container>`, `<Row>`, `<Col>`, `<Card>`, `<Nav>`

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Exercise 6 — Student REST API with Express + React Bootstrap Frontend

Topic: Express Routing

Scenario

Build a Student Management API with Express backend and React frontend.

Backend routes:

GET /students

POST /students

PUT /students/:id

DELETE /students/:id

Task

1. Create Express backend routes
2. Display students in Bootstrap table
3. Add new student using Bootstrap form
4. Edit student using **<Modal>**
5. Show success/error messages using **<Alert>**

State Management Requirement

Maintain:

- Student list
- Form data
- API response messages

React Bootstrap Components need to be used

- **<Form>**, **<Table>**, **<Button>**, **<Modal>**, **<Alert>**, **<Container>**